

ANALOG CMOS CIRCUIT DESIGN SS26 - PRACTICAL EXERCISE III

Two-Stage Operational Amplifier Frequency Analysis I

Tutors

Prof. Dr.-Ing. M. Kuhl (matthias.kuhl@imtek.de)

M.Sc. C. Grandauer (grandauer@imtek.de)

**M.Sc. Dikshith Gopala Krishna
(dikshith.gopalakrishna@imtek.uni-freiburg.de)**





Analog CMOS Circuit Design SS26

Practical Exercise III – Frequency Analysis I

Group

Name

Matriculation Number

Date

Feedback

.....

.....

.....

.....

.....

.....

.....

Points **Mark**

Date **Signature**

No review without this standard form!



ANALOG CMOS CIRCUIT DESIGN SS26

PRACTICAL EXERCISE III

An operational amplifier is mostly used in a feedback configuration as illustrated by the example of the buffer shown in **Fehler! Verweisquelle konnte nicht gefunden werden..** The purpose of this practical exercise is to highlight the importance of the frequency response and to show how to meet related specifications.

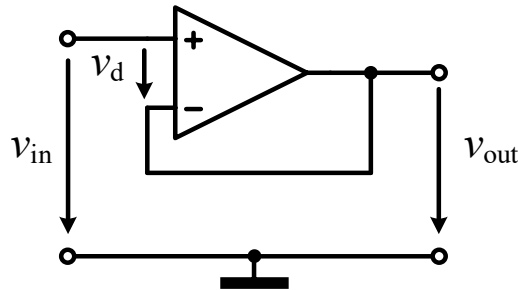


Figure 1: Buffer

T1: Negative Pole

- Calculate the magnitude and phase of the transfer function $TF_1(s) = \frac{A_1}{1 + s/|p|}$.
- Sketch the magnitude and phase for $A_1 = 100$ and $p = 10$ (Matlab, Mathematica).
- Consider your sketch. At which frequency does the pole approximately start (stop) to modify the magnitude and the phase?
- What is the value of the magnitude and the phase for $s = j\omega = jp$?
- Calculate the unity-gain frequency ω_u , i.e., $|TF_1(\omega_u)| = 1$. Simplify your result under the assumption that A_1 is very large.

T2: Positive Zero

- Calculate the magnitude and phase of the transfer function $TF_2(s) = A_2(1 - s/|z|)$.
- Sketch the magnitude and phase for $A_2 = 100$ and $z = 10$ (Matlab, Mathematica).
- Consider your sketch. At which frequency does the zero approximately start (stop) to modify the magnitude and the phase?



ANALOG CMOS CIRCUIT DESIGN SS26

PRACTICAL EXERCISE III

T3: Two-Stage Amplifier

Consider now the small signal gain of a generic amplifier given by

$$\frac{u_{out}}{u_{in}} = A(s) = \frac{A_{DC}}{\left(1 + \frac{s}{|p_1|}\right) \left(1 + \frac{s}{|p_2|}\right)} \left(1 - \frac{s}{|z_1|}\right)$$

- a) Apply your results of T1 and T2 in order to explain in a few words why the unity-gain frequency ω_u is determined by p_1 if we assume that $p_2 \geq 3\omega_u$ and $z_1 \geq 10\omega_u$. Determine ω_u in terms of A_{DC} and p_1 .
- b) Determine the location of p_2 in multiples of ω_u using your previous result so that the phase margin of $A(s)$ equals 45° , 60° and 75° while taking the zero z_1 , and both poles p_1 and p_2 into account.

Assumptions: $z_1 = 10\omega_u$, $A_{DC} \gg 1$

- c) Explain whether the phase margin will be increased or decreased if the zero moves from $10\omega_u$ to $5\omega_u$.

T4: Phase Margin & Feedback

Calculate the small signal transfer function v_{out}/v_{in} of the circuit shown in **Fehler! Verweisquelle konnte nicht gefunden werden.** while assuming a two-pole model for the operational amplifier, i.e.,

$$A(s) = \frac{A_{DC}}{\left(1 + \frac{s}{|p_1|}\right) \left(1 + \frac{s}{|p_2|}\right)}$$

Bonus Task

Simulate the step response of the circuit based on the calculated transfer function (Matlab) for $A_0 = 10000$, $p_2 \approx 2\pi 100$ and

$$p_1 \approx 2\pi 0.01414 \text{ (} 45^\circ \text{ phase margin)}$$

$$p_1 \approx 2\pi 0.00667 \text{ (} 60^\circ \text{ phase margin)}$$

$$p_1 \approx 2\pi 0.00179 \text{ (} 75^\circ \text{ phase margin)}$$



ANALOG CMOS CIRCUIT DESIGN SS26
PRACTICAL EXERCISE III

What can be observed?

